

Data Science & ML Assignment

Project: Predictive Analytics & Customer Segmentation

Phase 1: Exploratory Data Analysis (EDA)

Data Cleaning & Visualization

Objective: Understand the underlying patterns and quality of the provided dataset.

- Handle missing values and outliers using statistical imputation or removal.
- Perform Univariate and Bivariate analysis.
- Create a correlation heatmap to identify feature dependencies.

Phase 2: Feature Engineering & Preprocessing

Data Transformation

Prepare the data for algorithmic consumption.

- Encode categorical variables (One-Hot or Label Encoding).
- Scale numerical features using `StandardScaler` or `MinMaxScaler`.
- Perform Dimensionality Reduction using PCA if features exceed 20.

Phase 3: Machine Learning Modeling

Supervised & Unsupervised Learning

Task A: Classification/Regression model to predict target outcomes.

Task B: K-Means Clustering to segment customers based on behavior.

- Split data into Training, Validation, and Test sets (80/10/10).
- Implement a baseline model (e.g., Logistic Regression or Random Forest).
- Hyperparameter tuning using GridSearchCV.

Phase 4: Evaluation & Insights

Performance Metrics

- For Classification: Accuracy, Precision, Recall, and F1-Score.
- For Clustering: Silhouette Score and Elbow Method visualization.
- Provide a brief summary of the business impact based on model findings.

Recommended Tools: Python, Pandas, Scikit-Learn, Matplotlib/Seaborn, Jupyter Notebook.